CYS103 Linear Algebra (2026/09)
TUTORIAL 3

Chapter 2: Matrix Algebra

1.1 Matrix Operations
1.2 Inverse of a Matrix
1.3 Dimension and Rank

1. Let $A=\begin{bmatrix}2 & 0 & -1\\ 4 & -5 & 2\end{bmatrix}$, $B=\begin{bmatrix}7 & -5 & 1\\ 1 & -4 & -3\end{bmatrix}$, $C=\begin{bmatrix}1 & 2\\ -2 & 1\end{bmatrix}$, $D=\begin{bmatrix}3 & 5\\ -1 & 4\end{bmatrix}$, $E=\begin{bmatrix}-5\\ 3\end{bmatrix}$. Compute each matrix sum or product if it is defined.

   a. $-2A$, $B-2A$, AC, CD

   b. $A+2B$, $3C-E$, CB, EB

2. Let $A=\begin{bmatrix}1 & 1 & 1\\ 1 & 2 & 3\\ 1 & 4 & 5\end{bmatrix}$ and $D=\begin{bmatrix}2 & 0 & 0\\ 0 & 3 & 0\\ 0 & 0 & 5\end{bmatrix}$. Compute AD and DA. Determine whether the matrices $AD=DA$.

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3. Let $A = \begin{bmatrix} 2 & 5 \\ -3 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 4 & -5 \\ 3 & k \end{bmatrix}$. What value(s) of k, if any, will make $AB = BA$?

4. Let $\mathbf{u} = \begin{bmatrix} -2 \\ 3 \\ -4 \end{bmatrix}$ and $\mathbf{v} = \begin{bmatrix} a \\ b \\ c \end{bmatrix}$. Compute $\mathbf{u}^T\mathbf{v}$, $\mathbf{v}^T\mathbf{u}$, $\mathbf{u}\mathbf{v}^T$ and $\mathbf{v}\mathbf{u}^T$.

5. If $\mathbf{u}$ and $\mathbf{v}$ are in $\mathbb{R}^n$, how are $\mathbf{u}^T\mathbf{v}$ and $\mathbf{v}^T\mathbf{u}$ related? How are $\mathbf{u}\mathbf{v}^T$ and $\mathbf{v}\mathbf{u}^T$ related?

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6. Find the inverse of the matrix A:

$$A = \begin{bmatrix} 8 & 5 \\ -7 & -5 \end{bmatrix}$$

7. Solve the following system by using the inverse of the matrix.

$$\begin{aligned} 8x_1 + 6x_2 &= 2 \\ 5x_1 + 4x_2 &= -1 \end{aligned}$$

8. Use the algorithm to find the inverses of the following matrices.
   a) $A = \begin{bmatrix} 1 & 2 \\ 4 & 7 \end{bmatrix}$

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b) $B = \begin{bmatrix} 1 & 0 & -2 \\ -3 & 1 & 4 \\ 2 & -3 & 4 \end{bmatrix}$

9. Given that the following matrices are $n \times n$ matrices. Determine which of the matrices are invertible. (Hint: Use as few calculation steps as possible.)

a) $\begin{bmatrix} -1 & -3 & 0 & 1 \\ 3 & 5 & 8 & -3 \\ -2 & -6 & 3 & 2 \\ 0 & -1 & 2 & 1 \end{bmatrix}$

b) $\begin{bmatrix} 1 & -5 & -4 \\ 0 & 3 & 4 \\ -3 & 6 & 0 \end{bmatrix}$

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10. Let $\mathbf{v}_1 = \begin{bmatrix} 2 \\ 3 \\ -5 \end{bmatrix}, \mathbf{v}_2 = \begin{bmatrix} -4 \\ -5 \\ 8 \end{bmatrix}$ and $\mathbf{W} = \begin{bmatrix} 8 \\ 2 \\ -9 \end{bmatrix}$. Determine if $\mathbf{W}$ is in the subspace of $\mathbb{R}^3$ generated by $\mathbf{v}_1$ and $\mathbf{v}_2$.

11. State the dimensions of Nul A and Col A of the following coefficient matrices.

a) $A = \begin{bmatrix} 1 & -3 & 2 & -4 \\ -3 & 9 & -1 & 5 \\ 2 & -6 & 4 & -3 \\ -4 & 12 & 2 & 7 \end{bmatrix} \sim \begin{bmatrix} 1 & -3 & 2 & -4 \\ 0 & 0 & 5 & -7 \\ 0 & 0 & 0 & 5 \\ 0 & 0 & 0 & 0 \end{bmatrix}$

b) $A = \begin{bmatrix} 1 & -2 & 9 & 5 & 4 \\ 1 & -1 & 6 & 5 & -3 \\ -2 & 0 & -6 & 1 & -2 \\ 4 & 1 & 9 & 1 & -9 \end{bmatrix} \sim \begin{bmatrix} 1 & -2 & 9 & 5 & 4 \\ 0 & 1 & -3 & 0 & -7 \\ 0 & 0 & 0 & 1 & -2 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$

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Application:

The actual colour a viewer sees on a screen is influenced by the specific type and amount of phosphors on the screen. So, each computer screen manufacturer must convert between the (*R*, *G*, *B*) data and an international standard for colour, which uses three primary colours, called *X*, *Y*, and *Z*. A typical conversion for short-persistence phosphors is

$$\begin{bmatrix} 0.61 & 0.29 & 0.15 \\ 0.35 & 0.59 & 0.063 \\ 0.04 & 0.12 & 0.787 \end{bmatrix} \begin{bmatrix} R \\ G \\ B \end{bmatrix} = \begin{bmatrix} X \\ Y \\ Z \end{bmatrix}$$

A computer program will send a stream of colour information to the screen, using standard data (*X*, *Y*, *Z*). Find the equation that converts these data to the (*R*, *G*, *B*) data needed for the screen's electron gun.